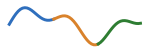

binary mask

(a)

HSI Cube & Ground Truth

H×W×B
labelled pixels
only

(b)

Stratified Pixel Split

16% train
4% val
80% test

(c)

Seg-PCA (Seg-3)

3 segments
10 PCs each
→ 30 features

(d)

mRMR Ranking

top 15
of 30
features

(e)

Metaheuristic Subset Search

PSO · GWO · GA
DE · MFO
5-NN fitness

(f)

Patch Extraction

25×25×K
around each
labelled pixel

(g)

Relevance-graded 3D/2D CNN

3 streams + CBAM
+ GAP + Softmax

all fitted on training pixels only (no test leakage)

Predicted class
(OA, AA, κ, F1)